

Caloric restriction and incidence of breast cancer.

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CONTEXT: Restricting caloric intake is one of the most effective ways to extend lifespan and to reduce spontaneous tumor occurrence in experimental animals, but whether similar associations hold in humans has not been appropriately studied. OBJECTIVE: To determine whether caloric restriction in early life reduces the risk of invasive breast cancer. DESIGN, SETTING, AND PARTICIPANTS: Retrospective cohort study using data from the Swedish Inpatient Registry, the Swedish Cancer Registry, the Swedish Death Registry, and the Swedish Fertility Registry. Participants were 7303 Swedish women hospitalized for anorexia nervosa prior to age 40 years between 1965 and 1998. Women were excluded (n = 31) if they were diagnosed with cancer prior to their first discharge from hospitalization for anorexia nervosa. MAIN OUTCOME MEASURE: Incidence of invasive breast cancer. RESULTS: Compared with the Swedish general population, women hospitalized for anorexia nervosa prior to age 40 years had a 53% (95% confidence interval [CI], 3%-81%) lower incidence of breast cancer; nulliparous women with anorexia nervosa had a 23% (95% CI, 79% higher to 75% lower) lower incidence, and parous women with anorexia nervosa had a 76% (95% CI, 13%-97%) lower incidence. CONCLUSIONS: Severe caloric restriction in humans may confer protection from invasive breast cancer. Low caloric intake prior to first birth followed by a subsequent pregnancy appears to be associated with an even more pronounced reduction in risk.